Microsoft Power Tools (Power Platform):

- **Power BI** – Business Intelligence & Data Visualization
- **Power Apps** – Low-Code App Development
- **Power Automate** – Workflow Automation
- **Power Virtual Agents** – Chatbot Creation
- **Power Pages** – Low-Code Website Development

ETL Cloud Platforms:

- **Azure Data Factory**
- **AWS Glue**
- **Google Cloud Dataflow**
- **Snowflake (with Snowpipe)**
- **Informatica Intelligent Cloud Services**
- **Talend Cloud Data Integration**
- **Matillion**
- **Fivetran**
- **Stitch**
- **Hevo Data**

Star and Snowflake schemas are two common ways of organizing data models.

In Power BI (and in data warehousing in general), Star and Snowflake schemas are two common ways of organizing data models. They define how fact tables and dimension tables are related for analysis.

1. Star Schema

Structure:

- ✓ A **central Fact Table** (holds numeric, measurable data like sales, revenue, quantity).
- ✓ **Dimension Tables** (hold descriptive attributes like customer name, product category, region).
- ✓ Dimensions **directly connect** to the fact table — no extra intermediate tables.

Looks like: A star shape — fact in the center, dimensions around it.

Advantages:

- ✓ Easy to understand and query.
- ✓ Faster performance (fewer joins).
- ✓ Best for reporting tools like Power BI.

Example:

```
    Customers
        |
Products - FactSales - Dates
        |
     Stores
```

2. Snowflake Schema

Structure:

- ✓ Similar to star schema, **but** dimension tables are **normalized** into multiple related tables.
- ✓ This means dimensions can have **sub-dimensions**.

Looks like: A snowflake — more branching because dimensions are split into more tables.

Advantages:

- ✓ Saves storage by avoiding duplicate data.
- ✓ Better for highly detailed, normalized databases.

Disadvantages:

- ✓ More complex queries (more joins → slower).
- ✓ Harder for non-technical users to understand.

Example:

```
      CustomerCity - Customers
                  |
ProductsCategory - Products - FactSales - Dates
                  |
         StoreRegion - Stores
```

Key Difference Table

Feature	Star Schema □	Snowflake Schema ❄ □
Data Redundancy	More redundancy	Less redundancy
Performance	Faster (fewer joins)	Slower (more joins)
Complexity	Simple structure	More complex structure

Feature	Star Schema □	Snowflake Schema ❄ □
Use Case	Reporting, analytics (Power BI)	Data warehouses with normalization

SharePoint

1. What is SharePoint?

- Microsoft's collaboration and document management platform.
- Used for file storage, team sites, and business workflows.
- Can host Excel, CSV, and other files that Power BI can connect to.
- Integrates well with Microsoft 365, Teams, and Power BI.

2. Using SharePoint in Power BI

a) As a Data Source

You can connect Power BI to:

- **SharePoint Document Library** – to fetch all files in a library.
- **SharePoint Folder** – to combine multiple files in the same folder.
- **SharePoint List** – to pull structured table-like data.

Steps to connect:

- Get Data → Select SharePoint Folder or SharePoint List.
- Enter the SharePoint site URL (not file link).
- Provide Microsoft credentials.
- Use Power Query to clean, transform, and load the data.

b) As a Report Hosting Platform

- You can publish Power BI reports to Power BI Service and then embed them into SharePoint Online pages.
- This allows your team to view reports without leaving SharePoint.

Steps to embed Power BI in SharePoint:

- In Power BI Service, open the report.
- Click File → Embed in SharePoint Online.
- Copy the link and paste it into a SharePoint Modern Page using the Power BI web part.

3. Advantages of SharePoint + Power BI Integration

- Centralized data storage & reporting access.
- Real-time collaboration on Excel/CSV files.
- Role-based security control.
- Works with Power Automate for workflow automation.

DirectQuery vs Import Mode in Power BI

1. Import Mode

Definition: Data is **copied and stored** inside the Power BI file (.pbix) during refresh. Queries are run on this **cached data**, not on the live source.

Key Points:

- **Performance:** Usually **faster** because data is pre-loaded in memory.
- **Refresh:** Needs manual or scheduled refresh to update data.
- **File Size:** Large datasets can make the .pbix file big.

- ❖ **Use Case:** Best for **small-to-medium datasets** where near-real-time data is not required.

Advantages:

- ❖ Fast report performance (in-memory analytics).
- ❖ Allows full use of Power BI features (complex DAX, transformations).
- ❖ Can work offline after import.

Limitations:

- ❖ Data can be outdated between refreshes.
- ❖ Large datasets may hit Power BI capacity
- ❖ limits (e.g., 1 GB for Pro, 10 GB for Premium per dataset).

2. DirectQuery Mode

Definition: Data stays in the **source database**. Queries are sent **live** to the source whenever a visual is refreshed.

Key Points:

- ❖ **Performance:** Depends on the **source system speed** and network latency.
- ❖ **Refresh:** Always live; no need to schedule refresh for updated data.
- ❖ **File Size:** Much smaller .pbix file since no data is stored locally.
- ❖ **Use Case:** Best for **large datasets** or scenarios where real-time data is required.

Advantages:

- ❖ Real-time or near-real-time analytics.
- ❖ No need to store huge datasets in Power BI.
- ❖ Can handle billions of rows (limited by source capability).

Limitations:

- ✓ Slower performance if source is slow or overloaded.
- ✓ Limited Power BI features compared to Import Mode (some DAX functions not available).
- ✓ Requires a reliable, high-speed connection to the data source.

Quick Comparison Table

Feature	Import Mode	DirectQuery
Data Storage	In Power BI (.pbix file)	In source database
Performance	Faster (in-memory)	Slower (depends on source)
Data Freshness	Needs scheduled refresh	Always live
File Size	Larger	Smaller
Feature Availability	Full Power BI functionality	Some features restricted
Use Case	Small/medium datasets, fast analytics	Very large datasets, real-time needs